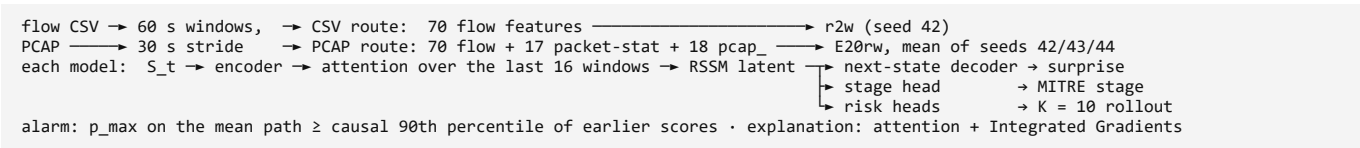


NetWM: Architecture

SIH 2026 · PS 26153 (NTRO) · AI based Network Attack Forecasting from Network Traffic Data

NetWM learns how a network's state evolves from one time window to the next, and rolls those dynamics forward to score the next five minutes. It is a world model with prediction heads, not a flow classifier. Every number below comes from results/; every design choice has its reason in decisions.md (D-xxx).



1. Network state s_t

- Windows.** Traffic is aggregated into 60 s windows at a 30 s stride. S_t is one feature vector per window, because consecutive flow records are not consecutive states (D-002).
- Flow features (70, both routes).** TCP flag ratios, the protocol and service mix, bytes, packets, duration and inter-arrival statistics, bidirectional ratios, unique IPs and ports, port entropy, fan-out, and ports per pair.
- Packet features (35, PCAP route).** 17 per-packet statistics a flow meter records, and 18 measured from the packets themselves: TTL, fragments, retransmissions, zero windows, the payload histogram, inter-packet timing, SYN-only and RST shares.
- PCAP converter.** A capture becomes flows by the corrected extraction's rules (D-033, D-043), traced flow by flow against the corrected CSVs.
- Scaling.** log1p on heavy tails, then a scaler fitted on training days only (D-014).
- Data.** The corrected CIC-IDS2017 re-extraction (D-001). Labels are ordered MITRE stages, with Impact off the progression axis (D-003); - Attempted traffic sets no stage (D-009).

2. World model

An RSSM-style latent model of about 0.6 M parameters (r2w: 584,470; E20rw: 593,500 each).

- An encoder, then causal attention over the last 16 windows. Each key is positioned by its distance from the query, so a window's score never depends on how much of the capture follows it (D-041).
- A stochastic latent with a learned prior: the transition function the rollouts imagine with.
- Heads on any real or imagined state: a next-state decoder (its error is the *surprise*), a MITRE stage head, and compromise, attack and escalation risk (D-016).
- Streaming.** The latent and the last 15 embeddings carry across calls. Scoring a feed in 60-window chunks equals one pass to 2.3e-10 on every served checkpoint (D-041, gate G1).
- What the dynamics earn.**
 - Against the same encoder predicting "within K" directly, the world model ranks pre-onset windows better (0.70 against 0.61) and detects better (PR-AUC 0.39 against 0.22) (E24).
 - Its rollouts beat "nothing changes" averaged over steps 2-10, though not at k = 1 (E10).
 - Logistic regression ranks pre-onset windows as well (0.66 against 0.65, E26). Anticipation as a ranking does not need a world model; detection at a usable alarm cost does.

3. Forecast, alarm and explanation

- Forecast.** From each window's filtered state the model imagines K = 10 steps (5 minutes, D-013). The alarm score p_max is the largest compromise risk over the rollout, on the deterministic mean path (D-019); Monte-Carlo rollouts give the uncertainty band.
- Alarm.** A causal alert budget: each window's threshold is the 90th percentile of the scores before it, with no alarm in the first 10 minutes (D-034). Thresholds tuned on training days fire nothing on a held-out day (D-017).
- Stage.** The stage head maps each forecast to MITRE ATT&CK (src/netwm/labels/mitre_map.py).
- Explanation.** When: attention over earlier windows. What: Integrated Gradients over the input features on alarm windows (D-018); the logistic-regression baseline gets exact linear SHAP.

4. Served routes

input	model	inputs	why
flow CSV	r2w, seed 42 (models/n8-r2w-s42/)	70	passed D-041's gate G3
PCAP	E20rw, mean of seeds 42/43/44 (models/m1v2-n8-e20rw-s4{2,3,4}/)	105	D-038's aggregation; gate G2

- A PCAP never falls back to the CSV model, and a CSV never reaches the packet model.
- The PCAP route's live scores follow its training-state scores at 0.991 on a real capture (0.104 before D-043).
- Deployment.** A FastAPI backend and a static dashboard, fully offline: a test makes every outbound connection raise. The 17 served checkpoint files are in the repository (D-046), identified by SHA-256.
- Frozen evidence.** Tag submission-freeze (results/runs/submission-freeze/manifest.json).

5. Results: CIC-IDS2017, leave-one-day-out, causal threshold

held-out Thursday	PR-AUC	F1	precision	recall	FPR
CSV route: r2w seed 42 (seeds 43 / 44: 0.340 / 0.408)	0.677	0.591	0.574	0.608	0.093
PCAP route: E20rw mean (seeds: 0.409 / 0.542 / 0.726)	0.514	0.583	0.581	0.584	0.087
logistic regression, same features and threshold	0.139	0.112	0.167	0.084	0.087

- **Friday** holds only a C2 family seen on no training day: both routes are near floor (0.118, 0.137).
- **Packets help detection, not anticipation.** They raised Thursday PR-AUC from 0.43 to 0.56 over flow-only (E20r), but not the pre-onset ranking.
- **Anticipation (PCAP route, 19 onsets).** Pre-onset windows rank above background at 0.675 (chance 0.5). Alarm precision is 0.350 at FPR 0.129. Only 5 of 19 onsets are warned early ($p = 0.53$ against a circular-shift null).
- **Cross-family, not served** (each family held out in turn, 3 seeds, 95 % block-bootstrap intervals):
 - **CTU-13 (E25b).** Detection transfers firmly to NSIS (0.98). The model is inverted on Murlo and Rbot s11, scoring the bot below background. Most inversions are feature-direction flips (N-6a).
 - **CIC-IDS2018 (E28).** Held-out attack families are recognised:
 - BruteForce 0.89-0.99, DoS 0.81-0.93;
 - DDoS and Web partial (Web 0.64-0.77).
 - **Compromise does not carry** on CIC-IDS2018: Infiltration is inverted on one day (0.26-0.42), and Botnet (0.69-0.78) sits below LR (0.82).

6. Limitations

1. **No validated early warning** on any of the three datasets. The claim is "ranks pre-attack windows above background" (D-021).
2. **Compromise does not transfer to an unseen compromise family**, although attack families do (E25b, E28). The architecture stays as it is; no per-host or graph model was built (D-045).
3. **Risk scores, not calibrated probabilities** (E17): hence the alert budget.
4. **Explanations and stage labels are weak on unseen attacks.**
 - The top-8 attributions contain the known signature on only 1 of 6 held-out episodes (E11).
 - The stage head cannot recall a stage its fold never trained on (E9).
5. **Live-PCAP quality is not measured.** The PCAP converter's parity is substantially validated, but not exact: active_mean is still 9 % off (N-9). No live-PCAP detection number is claimed.